Astana IT University

Business Administration

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Weekly Records

(Weeks 3-5)

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Company A: Shell

Company B: KazMunayGaz(KMG)

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Weekly Record 3: Strategic Environment and Transfer Analysis

Concept

Key terms: PESTEL analysis, Strategic transfer analysis, Adopt, Adapt, Pilot, Delay, Reject, Methane Flaring

Question: Why does uncritical transfer of Shell's strategy to KMG guarantee failure?

Answer: Shell operates as an international capital allocator focused on maximizing free cash flow per share and providing private shareholders with high-quality assets. In contrast, KazMunayGas is a sovereign energy holding company in Kazakhstan, which has dual obligations: to ensure national fuel security, supply domestic oil refineries at subsidized prices, and maintain employment for more than 52,000 workers in vulnerable regions such as Mangystau and Atyrau. Repeating Shell's strategy of aggressively divesting assets and switching to market-based fuel prices would trigger severe domestic inflation, an energy shortage, and regional socio-economic instability.

Application – Macro-Environment and Opportunity map

Sector	Kazakhstan Operational Reality	Strategic impact on KMG	Digital and AI Opportunity Idea
Political / Legal	A government directive requiring the supply of crude oil to domestic oil refineries (ANPZ, PNHZ, PKOP) at subsidized prices.	It limits the profit margin compared to the ratio of net export income.	AI-driven performance optimization between export routes and domestic oil refineries.
Economic	Financial indicators for 2025: revenue - 9,371 billion tenge, EBITDA - 2,393 billion tenge, net debt/EBITDA ratio - 0.16 times higher, taxes - 2.3 trillion tenge.	A good balance, but capital investments must be strictly allocated according to priorities.	Predictive financial modeling and stress testing of cash flows under conditions of Brent crude oil price volatility.
Social	The largest employer in the region (>52,000 employees) there is acute social concern about job losses in Mangystau.	The number of staff cannot be drastically reduced without causing social unrest.	Safety monitoring using TUMAR computer vision to protect frontline workers and VR/AI training.
Technological	Implementation of the proprietary ABAI Big Data/AI platform and drilling of thin wells.	The outdated infrastructure requires a transition to preventive maintenance.	Scaling of ABAI Waterflooding AI (1,660 injection cells) and failure prediction algorithms.

Environmental	Low-carbon program until 2060. 98.8 % of associated petroleum gas utilization; hybrid power plant in Zhanaozen with a capacity of 247 MW.	International lenders are putting pressure on ESG, demanding a reduction in the intensity of work within categories 1 and 2.	Satellite and optical gas imaging (OGI) with leak detection functionality using artificial intelligence for OGMP 2.0 Methane Atlas.
Global and Trade	Export dependence on the Caspian Pipeline Consortium (CPC, capacity 83 million tons).	Geopolitical risks associated with the bottleneck in the Black Sea require strict control over transit.	Optimizing supply chain routes using artificial intelligence and predictive planning of vessel movements with KMG Maritime.

Verification and AI Record

KMG Financials: 2025 Revenue (9,371B KZT) and EBITDA (2,393B KZT) verified via KMG 2025 Consolidated Financial Statements, Note 6, p. 326.
Net Debt/EBITDA: 0.16x ratio verified via KMG 2025 Annual Report, p. 30.
ABAI Platform: Waterflooding coverage across 1,660 injection cells verified via KMG 2025 Annual Report, p. 178.
Shell Benchmarks: Liquids output (1.4M bpd) & Scope 1&2 reductions (36%) verified via Shell Annual Report 2025, p. 385.
Information about the use of AI: ChatGPT and Gemini were used only to check grammar and errors, if necessary, for translation. All content, analysis, and quantitative indicators were independently prepared and verified using primary sources.

1. Adopt: AI Predictive Maintenance and Computer Vision Safety Systems
   Reasoning: KMG should fully adopt Shell's practice of enterprise-wide predictive failure analytics and CCTV computer vision safety monitoring (modeled on Shell's T-Pulse). Extending TUMAR and ABAI across all domestic refineries (ANPZ, PNHZ, PKOP) will eliminate unplanned downtime and drastically reduce industrial accidents [3], [4].
2. Adapt: Climate Strategy and Decarbonization Framework
   Reasoning: Shell's Net-Zero framework must be adapted to Kazakhstan's fossil-reliant economy. KMG should focus on Scope 1 and 2 methane abatement (OGMP 2.0 Gold Standard), hybrid renewable/gas generation (Zhanaozen 247 MW), and CCUS screening at gas processing units rather than divesting producing hydrocarbon assets [3], [4], [5].
3. Pilot: Digital Twin Reservoirs and Advanced EOR / Slim Hole Drilling
   Reasoning: Shell's digital twin subsoil modeling and advanced EOR techniques should be piloted across declining mature fields (Embamunaygas, Ozenmunaygas) to optimize sweep efficiency and extend field life [3], [4].

4. Delay: Mass Capital Allocation to Green Hydrogen and EV Retail Infrastructure
Reasoning: Unlike Shell (which expands Shell Recharge and green hydrogen electrolysers), KMG should delay heavy CAPEX in green hydrogen and EV charging networks until EV fleet adoption and market demand mature in Central Asia, restricting current involvement to R&D desk studies [3], [4], [5].
5. Reject: Pure Market Pricing and Aggressive Divestment of Social Downstream Assets
Reasoning: Shell's strategy of shutting down or selling lower-margin refining assets must be explicitly rejected. Abandoning domestic fuel subsidies or selling domestic refineries would cause severe fuel shortages, economic inflation, and social disruption in Kazakhstan [1], [3], [5], [9].

5. Reference List of wr3:

[1] OpenStax, Introduction to Business 2e, Rice University, Houston, TX, USA, 2024. [Online]. Available: https://openstax.org
[2] KAZENERGY Association, National Energy Report 2023, Astana, Kazakhstan, 2023.
[3] JSC NC "KazMunayGas", Annual Report 2025: Global Standards. National Priorities, Astana, Kazakhstan, Apr. 28, 2026. [Online]. Available: https://www.kmg.kz
[4] Shell plc, Annual Report and Accounts 2025: Deliver More Value with Less Emissions, London, UK, Mar. 11, 2026. [Online]. Available: https://www.shell.com
[5] JSC NC "KazMunayGas", Consolidated Financial Statements for the Year Ended December 31, 2025, PwC Audit Report, Astana, Kazakhstan, Mar. 5, 2026.
[6] E. Akkenzhenov, Ministry of Energy Report on 2025 Performance and 2026 Digitalization Objectives, Ministry of Energy RK, Astana, 2026.
[7] KMG International N.V., KMGI Financial Performance and European Refining Operational Brief, Press Release, Bucharest, Romania, July 29, 2026.
[8] JSC NC "KazMunayGas", Corporate Governance and Shareholder Structure, Official Portal, 2026. [Online]. Available: https://www.kmg.kz
[9] Ministry of Energy of the Republic of Kazakhstan, Digitalization of Energy Sector and Refining Outlook, Official Report, Astana, 2026.
[10] Caspian Pipeline Consortium (CPC), Operational Crude Oil Transportation Volumes Brief, 2026.
[11] Administration of the President of the Republic of Kazakhstan, Address of the Head of State Kassym-Jomart Tokayev on AI and Digital Transformation, Official Site of the President of RK, 2026.

6.Capstone Link

Developed into Capstone Section 3: Strategic Environment and AI Opportunity Map

Weekly record 4: Human-AI Operating model

Concept

Key Terms: POLC Framework (Planning, Organizing, Leading, Controlling), Span of Control, Centralization vs. Decentralization, 4 Allocation Categories (Human Only, Human + AI, AI Executes + Human Review, Automated).

Question: Why can AI execute a task but not be accountable for it?
Answer: Accountability requires legal authority, moral agency, and financial liability. Under the Kazakhstan Labor Code and corporate governance standards, an AI algorithm cannot hold fiduciary duty, sign statutory safety approvals, or be held legally liable for industrial accidents or employment decisions. AI processes data and executes rules, but a named human manager must remain legally accountable for the outcome.

Application - Human-AI Task Allocation Matrix (KMG)

#	Function	KMG Operational Task	Category	Justification (Test + Fact)	Accountable Role	Review
1	Planning	Allocate annual CAPEX across regional operating assets	Human + AI	Cost of Error - Strategic capital decisions impact 0.16x leverage; AI models scenarios, human decides.	CFO	☐
2	Planning	Forecast domestic fuel demand for ANPZ/PNHZ allocation	AI Executes + Review	Volume and Rate - AI analyzes historical regional consumption data across 14 regions.	Head Marketing	☐
3	Organizing	Restructure workforce in Mangystau operating assets	Human Only	Accountability and Empathy - Affects >52,000 workers' livelihoods and regional stability.	HR Director	☐
4	Organizing	Schedule routine refinery valve & pump inspections	Automated	Volume and rate - Fixed calendar and sensor thresholds; fallback to manual logbook.	Maintenance Lead	☐
5	Leading	Conduct executive performance & dismissal reviews	Human Only	Accountability - Legal employment rights under KZ Labor Code.	Board / MD	☐

6	Leading	Monitor workplace safety violations on workover crews	AI Executes + Review	Cost of Error and Verification - TUMAR computer vision flags breaches; inspector verifies.	Chief HSE	☑
7	Controlling	Monitor predictive equipment failure at refineries	AI Executes + Review	Cost of Error - Prevents unplanned downtime; certified engineer verifies anomaly alerts.	Chief Engineer	☑
8	Controlling	Adjust waterflooding injection rates across 1,660 cells	AI Executes + Review	Volume & Rules - ABAI AI optimizes sweep efficiency; engineer approves changes.	Reservoir Eng.	☐

Organizational Design Decision: Centralized ownership by the KMG Digital Transformation Department for AI platform architecture (ABAI/TUMAR), coupled with decentralized line-management accountability where operational VPs and Refinery Chief Engineers retain sole legal signing authority.

Challenge Round Response: Following group critique, Task 8 (ABAI waterflooding) was maintained as 'AI Executes + Human Review' rather than 'Automated' because reservoir pressure miscalculations can permanently damage mature subsoil formations.

Verification + AI Record

All task specifications and system names (TUMAR, ABAI, 1,660 cells) were verified against the KMG 2025 Annual Report (Digitalization Section, pp. 178–181). Level 2 AI usage was limited to table formatting and verification syntax also for translation.

Transfer Analysis

KMG adopts Shell's computer vision safety monitoring (TUMAR) and predictive failure analytics but rejects AI-driven automated workforce downsizing due to sovereign social obligations.

Evidence (IEEE References)

[1] KMG Engineering, ABAI System Operational Brief, Astana, 2026.

[2] Wendover Productions, How Boeing Lost Its Way, YouTube, 2024.

[3] Klarna, AI Assistant Operational Release & Quality Calibration Review, Press Release, 2024-2025.

Capstone Link

Developed into Capstone Section 4: Human-AI Operating Model.

Weekly record 5: Operations and process control

Concept

Key Terms: Production Planning, Process Types, Just-In-Time (JIT), Critical Path, Total Quality Management (TQM), Lean Manufacturing, Bottleneck, Key Performance Indicator (KPI), Human Review Point, Escalation Path, Manual Fallback.

Question: Why does automating a non-bottleneck does not increase overall output?
Answer: Total process throughput is governed strictly by the capacity of the bottleneck (the constraint). Automating a non-bottleneck step increases processing speed at that specific station but merely accelerates the accumulation of Work-In-Progress (WIP) inventory upstream of the bottleneck without increasing the final rate of completed output.

Application - KMG Process Mapping & Controls

A. Paper Airplane Factory Simulation Debrief: Transitioning from Round 1 (Batching & Push) to Round 2 (One-Piece Flow & Pull) eliminated WIP accumulation on desks, reduced defect rates by inspecting at each station, and cut total lead time to the first accepted unit by over 60%.

B. Company B Selected Process: Refining Equipment Maintenance Workflow at Atyrau Refinery (ANPZ).
Trigger: Continuous vibration and thermal sensor telemetry from critical processing units.
Result: Elimination of unplanned refinery outages and transition to 3-year extended turnaround cycles.

C. Current-State Process Map

#	Actor (Swimlane)	Step (Verb + Object)	Time / Delay	Problem Observed & Bottleneck
1	Field Operator	Perform physical inspection walk-throughs	2 hrs / shift	Manual logging, delayed anomaly detection.
2	Local Gauge	Record temperature & vibration numbers	1 hour	Data silos, paper-based records.
3	Maintenance Staff	Analyze historical failure trends in Excel	12-24 hours	BOTTLENECK: Reactive maintenance; equipment fails before review.
4	Chief Engineer	Approve emergency repair work orders	4 hours	High emergency maintenance OPEX.
5	Repair Crew	Shut down unit and execute repairs	48-72 hours	Unplanned refinery downtime (120 hrs/yr baseline).

D. KPI Definition: Unplanned Refinery Downtime (Hours per Year) & Maintenance OPEX. Baseline: 120 hours/year. Target: 15% reduction in unplanned downtime (<102 hrs/yr) and 10% reduction in maintenance OPEX by end of 2027.

E. Future-State Process Map (AI-Integrated)

#	Actor (Swimlane)	Step (Verb + Object)	Allocation Category	What Changed & Effect on KPI
1	IoT Sensors	Stream real-time telemetry	Automated	Replaced manual physical logging; continuous telemetry feed.
2	Predictive AI	Process telemetry and flag failure anomalies	AI Executes + Review	Eliminates reactive maintenance bottleneck via 14-day advance prediction.
3	Chief Engineer	Verify anomaly flag & approve work order	Human + AI	Human Review Point: Engineer checks AI diagnosis; prevents false shutdowns.
4	Automated System	Schedule parts procurement & crew dispatch	Automated	ERP automatically reserves spares; cuts planning lead time from 24h to 30m.
5	Maintenance Crew	Execute planned preventative maintenance	Human Only	Achieves 15% downtime reduction KPI through scheduled maintenance windows.

F. Operational Controls (Rubric Row 7):
Human Review Point: Certified Chief Mechanical Engineer must independently inspect AI diagnostic telemetry and physically sign off before any production unit is shut down.

Escalation Path: Anomaly flagged by AI, Senior Maintenance Tech notified within 15 mins, Chief Mechanical Engineer alerted within 1 hr, General Director notified within 4 hrs if estimated repair time exceeds 24 hrs.
Manual Fallback: If IoT sensor networks or AI predictive servers fail, plant operations immediately revert to manual hourly physical operator logbooks, analog pressure gauge checks, and standard physical maintenance schedules.

Fields 3-5 Synthesis Status: Absorbed directly into Assignment 2 Video Essay (Script, Evidence Folder, and AI-Use Declaration).

Capstone Link

Develops into Capstone Section 5a: Operations and Supply Chain.

CONSOLIDATED EVIDENCE FOLDER (IEEE REFERENCES)

[1] OpenStax, Introduction to Business 2e, Houston, TX, USA: Rice University, 2024. [Online]. Available: https://openstax.org

[2] KAZENERGY Association, National Energy Report 2023, Astana, Kazakhstan, 2023.

[3] JSC NC "KazMunayGas", Annual Report 2025: Global Standards. National Priorities, Astana, Kazakhstan, Apr. 28, 2026. [Online]. Available: https://www.kmg.kz

[4] Shell plc, Annual Report and Accounts 2025: Deliver More Value with Less Emissions, London, UK, Mar. 11, 2026. [Online]. Available: https://www.shell.com

[5] JSC NC "KazMunayGas", Consolidated Financial Statements for the Year Ended December 31, 2025, PwC Audit Report, Astana, Kazakhstan, Mar. 5, 2026.

[6] Wendover Productions, How Boeing Lost Its Way, YouTube, 2024. [Online]. Available: https://www.youtube.com/watch?v=URoVKPVDKPU

[7] Klarna, Klarna AI assistant handles two-thirds of customer service chats in its first month, press release, Feb. 27, 2024.

[8] Ministry of Energy of the Republic of Kazakhstan, Digitalization of Energy Sector and Refining Outlook, Official Report, Astana, 2026.

[9] Caspian Pipeline Consortium (CPC), Operational Crude Oil Transportation Volumes Brief, 2026.

[10] Lean Smarts, Lean Manufacturing One Piece Flow vs. Mass Production Paper Airplane Simulation, YouTube, 2022.